

Evidence before build

A work order earns an agent turn only after scope, evidence, safety, workspace, and cost boundaries are visible.

The flow moves from a bounded work order through a deterministic evidence-and-safety preflight to one of three explicit outcomes: proceed, narrow the work, or request human review. The work order remains the accountability boundary in both directions.

MERLIN SOFTWARE FACTORY - PUBLIC TEACHING EDITION

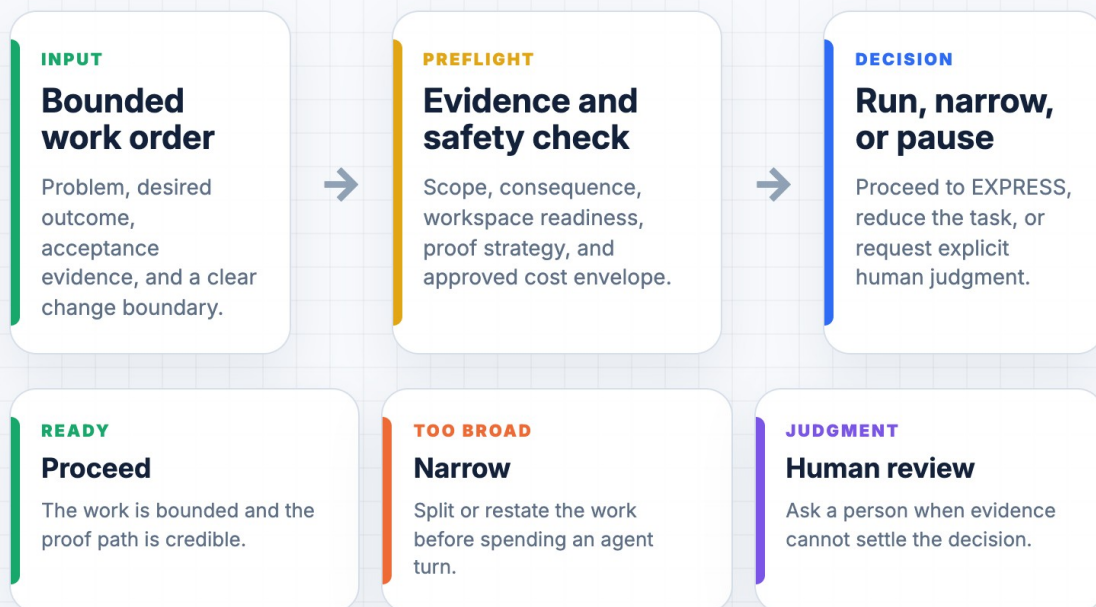
1/5

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THE FIRST DECISION IS WHETHER TO AUTOMATE

Preflight is deterministic wherever possible. It does not pretend that missing product judgment can be replaced by more agent activity.



The work order is the accountability boundary.

It carries intent into execution and carries evidence, limitations, cost, and next action back out.

One agent. Six disciplined phases.

The EXPRESS delivery arc preserves intent by keeping one accountable agent responsible from understanding through self-audit.

The six phases are Understand, Investigate, Plan, Test, Implement, and Self-audit. Continuity protects design intent and local context, but it does not replace deterministic checks, independent verification, cost boundaries, or human judgment.

THE EXPRESS DELIVERY ARC

2/5

One agent. Six disciplined phases.

Continuity preserves intent: the same accountable agent investigates, plans, proves, changes, and audits the work.

A REASONING ARC, NOT SIX AGENT HANDOFFS

Each phase produces evidence for the next. Ownership stays continuous while independent controls remain outside the agent's self-assessment.

1 INTENT

Understand

Restate the outcome, constraints, and acceptance conditions.

2 EVIDENCE

Investigate

Read the relevant system before proposing a change.

3 BOUNDARY

Plan

Declare the change surface, risks, and proof strategy.

4 PROOF FIRST

Test

Make expected behavior executable where the work allows it.

5 CHANGE

Implement

Make the smallest coherent change inside the declared scope.

6 CHALLENGE

Self-audit

Test the claim, record limitations, and prepare a clean handoff.

What continuity protects

Design intent, test reasoning, local context, and responsibility for the final claim.

What continuity does not replace

Deterministic checks, independent verification, cost boundaries, or human judgment.

Proof is not one undifferentiated gate

Verification is classified by consequence so rigor increases trust without turning every signal into critical-path ceremony.

Blocking checks stop unsafe work; conditional checks become blocking when relevant; advisory checks inform judgment; asynchronous checks improve later work. Independent challenge and bounded repair keep the implementer from being the only judge of its own claim.

VERIFICATION BY CONSEQUENCE

3/5

Proof is not one undifferentiated gate

The factory classifies checks by consequence so rigor increases trust without turning every signal into critical-path ceremony.

THE POLICY DECIDES WHAT FAILURE MEANS

Classification keeps safety strict, context honest, and learning out of the delivery bottleneck.

STOP

Blocking

Failure makes the result unsafe or untrustworthy. Stop the run or enter bounded repair.

executable proof

critical security

repository integrity

WHEN APPLICABLE

Conditional

The check becomes blocking when the work declares that concern or touches that surface.

user experience

performance

declared constraints

INFORM JUDGMENT

Advisory

The signal improves review and calibration without manufacturing unnecessary delivery friction.

risk framing

telemetry quality

documentation

AFTER DELIVERY

Asynchronous

The work improves future runs but should not hold the current result hostage.

retrospective

memory

improvement

INDEPENDENT CHALLENGE

The implementer is not the only judge

Deterministic evidence and a separate reading challenge overclaims before trust is granted.

BOUNDED REPAIR

Failure has a ceiling

Repair loops stop when progress stalls, cost rises, or judgment becomes necessary.

Automation earns trust by showing its work

State, evidence, cost, failures, and next actions remain visible so a person can intervene without reconstructing the run.

The agent produces evidence, the control plane keeps work convergent, and the operator decides with context. Durable state, observability, cost safety, and workspace isolation make human review a healthy architectural outcome rather than a hidden failure.

OPERATIONS AND HUMAN JUDGMENT

4/5

Automation earns trust by showing its work

State, evidence, cost, failures, and next actions stay visible so a person can intervene without reconstructing the run.

HUMAN REVIEW IS AN ARCHITECTURAL OUTCOME

The system does not hide uncertainty behind a success label. It turns unresolved judgment into a clean, evidence-rich handoff.

AGENT

Produce evidence

Scope, changes, tests, limitations, and the claimed outcome travel together.

intent

proof

limits

CONTROL

Keep state convergent

Resume recoverable work, detect repeated failure, and halt safely at explicit boundaries.

recovery

caps

diagnosis

OPERATOR

Decide with context

Approve, reject, narrow, or continue with the evidence and next action already assembled.

review

approval

handoff

DURABLE STATE

Work survives process boundaries

The work order, not an agent's private memory, remains the source of truth.

OBSERVABILITY

Outcome beats activity

Delivery, quality, cost, recovery, and adoption show whether the work actually holds.

COST SAFETY

No invisible runaway

Hard caps, no-progress detection, and approval boundaries stop unattended escalation.

WORKSPACE SAFETY

One bounded place to act

Each work order receives an isolated change surface and a recoverable trail.

A clean human handoff is a successful terminal state when product, risk, or operational judgment exceeds automation.

Every run can improve the next

Learning compounds only when it is verified, traceable, measured, and kept outside the delivery path whenever possible.

Outcome signals become verified memory, which can propose a bounded improvement. Each candidate must be evaluated and either promoted, discarded, or rolled back. The reusable rules are to bound work, preserve accountability, classify proof, escalate judgment, expose state and cost, and promote learning only after evidence.

MEMORY AND CONTINUOUS IMPROVEMENT

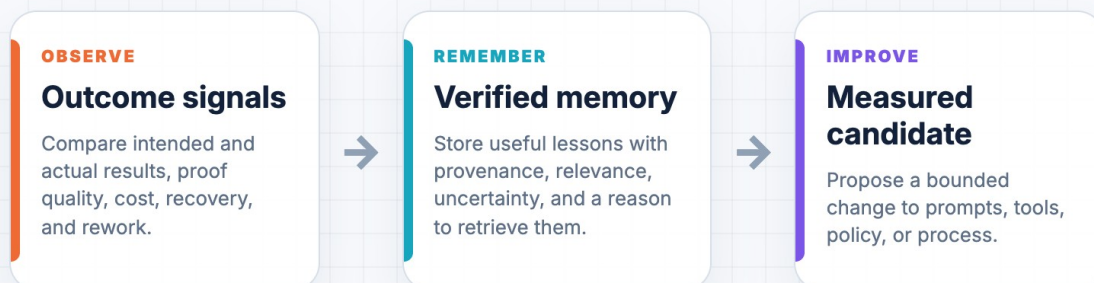
5/5

Every run can improve the next

Learning compounds only when it is verified, traceable, measured, and kept outside the critical delivery path whenever possible.

LEARNING FOLLOWS EVIDENCE

The factory records what happened, preserves provenance, checks remembered claims against current reality, and tests proposed improvements before promotion.



Improvement must earn promotion

Evaluate

Test against representative work with explicit success and regression criteria.

Promote

Adopt only when measured evidence shows a durable gain.

Discard or roll back

Reject variants that add risk, cost, or complexity without enough value.

- 1 Bound the work before execution.
- 2 Keep one agent accountable for the arc.
- 3 Classify proof by consequence.
- 4 Escalate judgment instead of inventing certainty.
- 5 Make state, cost, and recovery observable.
- 6 Promote learning only after measured evidence.